

F630 — NEW ROW (charge quantization) opens with a WIN: all four SM electric charges $\{u:+2/3, d:-1/3, e:-1, \nu:0\}$ are DERIVED, exactly and target-innocently, from four BST-derived inputs — the $SU(2)_L$ doublet splitting ($Q_{up}-Q_{down}=1$), generation neutrality (the 16 is anomaly-free $\rightarrow \Sigma Q=0$), the chargeless neutrino ($Q_\nu=0$), and $N_c=3$. The fractional $1/3$ IS $1/N_c$, forced by N_c *colored* quarks + neutrality — which is exactly Casey’s bulk(colored quark)-vs-boundary(colorless lepton) confinement structure. Discrete, exact, structural — BST’s strength, and the right pivot off the flavor swamp.

Lyra, Tue 2026-07-21. Pivot landing. After the flavor arc closed honestly (mechanism + unification derived, spectrum/addresses not), the pivot to charge quantization — discrete and structural — lands a clean derivation on the first pull.

The derivation (target-innocent — the charges are PRODUCED, not fit)

Four inputs, each BST-derived: 1. **** $SU(2)_L$ doublet splitting:**** within a doublet, $Q_{up} - Q_{down} = T_{3_up} - T_{3_down} = 1$ ($Sp(1)_L$, $T_3=\pm 1/2$). Holds for both the quark (u,d) and lepton (ν,e) doublets. 2. Generation neutrality: the fermion content is 16 = one $SO(10)$ spinor = anomaly-free, so the total electric charge per generation vanishes: $N_c(Q_u+Q_d) + (Q_\nu+Q_e) = 0$. 3. Chargeless neutrino: $Q_\nu = 0$ (the ν sits at the chargeless S^4 locus, F582/F583). 4. $N_c = 3$ (the root multiplicity, F579 — derived, not assumed).

Solve: - Leptons: $Q_\nu - Q_e = 1$ (doublet) with $Q_\nu=0 \Rightarrow Q_e = -1$. - Neutrality: $N_c(Q_u+Q_d) = -(Q_\nu+Q_e) = 1 \Rightarrow Q_u+Q_d = 1/N_c = 1/3$. - Quarks: $Q_u - Q_d = 1$ (doublet) with $Q_u+Q_d = 1/3 \Rightarrow Q_u = +2/3, Q_d = -1/3$.

$$\{Q_u, Q_d, Q_e, Q_\nu\} = \left\{+\frac{2}{3}, -\frac{1}{3}, -1, 0\right\} \quad \text{— exactly the SM charges, produced.}$$

Charge-sum check: $N_c(Q_u+Q_d) + (Q_\nu+Q_e) = 3 \cdot (1/3) + (0-1) = 0 \checkmark$.

★ The clean handle — the $1/3$ IS $1/N_c$ (bulk-vs-boundary)

The fractional quark charge comes in units of $1/N_c = 1/3$, and the $1/3$ is *literally* $1/N_c$ — forced by there being N_c *colored* quarks whose charge must sum ($\times N_c$) against the integer lepton charge to give zero. This is exactly Casey’s confinement geometry: - Quarks = colored BULK states $\rightarrow$ their charge is quantized in units of $1/N_c$ (the color dimension in the bulk fractionalizes the charge). - Leptons = colorless BOUNDARY (Shilov) states $\rightarrow$ integer charge. So “why are quark charges thirds and lepton charges integers” = “quarks are colored bulk (N_c), leptons are colorless boundary” — the bulk-vs-boundary split is the thirds-vs-integers split. Target-innocent, discrete, exact.

Why this is a derivation (not the standard SM assumption)

Charge quantization from anomaly cancellation is known in the SM — but BST *derives every input*: - $N_c=3$ = root multiplicity (F579) $\rightarrow$ the $1/3$ is *geometric*, not put in. - Neutrality = the 16 (one $SO(10)$ spinor) is anomaly-free $\rightarrow$ derived from the fermion content (which BST derives, F582). - The doublet splitting = $Sp(1)_L$ (derived weak sector). - $Q_\nu=0$ = the chargeless-locus neutrino (F582/F583, derived). And the bulk-vs-boundary geometric reading (color in the bulk fractionalizes; colorless boundary is integer) is the new structural content — it *explains* the fractionalization rather than assuming it. Tier: DERIVED (exact charges from derived inputs), target-innocent.

Row targets (next, all discrete/structural)

1. [DONE, this note] The $1/N_c$ fractionalization + the four charge values. Clean.
2. Hypercharge $Y = T_{3_R} + (B-L)/2$ (F582) $\rightarrow$ the full hypercharge assignments $\{Q_L: 1/6, u_R: 2/3, d_R: -1/3, L: -1/2, e_R: -1\}$ — check they all fall out of the same bulk-boundary + neutrality structure. (Likely needs the $T_{3_R}/(B-L)$ structure of F582, more than bulk-boundary alone — flag honestly.)
3. Anomaly cancellation as geometric: the full set of anomaly conditions ($[SU(3)]^2U(1)$, $[SU(2)]^2U(1)$, $U(1)^3$, $U(1)\text{-grav}^2$) — do they all reduce to the 16 being one spinor + $N_c=3$? (The 16 is automatically anomaly-free, so this is likely a restatement — bank the electric-charge-sum=0 as the clean one, don't over-claim the full anomaly structure.)

Tiers / handoffs

- All four SM electric charges DERIVED (doublet + neutrality + chargeless- ν + $N_c=3$), exact, target-innocent. The $1/3 = 1/N_c$ fractionalization = bulk(colored quark)-vs-boundary(colorless lepton). Clean pivot win.
- @Cal — target-innocence: the charges are *produced* from four BST-derived inputs (not fit); the $1/3 = 1/N_c$ with N_c derived (root multiplicity); the bulk-boundary reading explains the fractionalization. Standard-SM caveat: charge-quantization-from-anomaly is known — BST's addition is that every input is derived + the geometric bulk-boundary reason. Gate: don't over-claim the *full* hypercharge/anomaly structure yet (target 2/3) — bank the clean electric-charge derivation first.
- @Elie — verify: $\{u:+2/3,d:-1/3,e:-1,\nu:0\}$ from $N_c(Q_u+Q_d)+(Q_\nu+Q_e)=0$ + doublet splittings + $Q_\nu=0$ (target-innocent, produces the charges); the $1/N_c$ fractionalization. Then target 2 (full hypercharge $Y=T_{3_R}+(B-L)/2$, F582) — does it fall out of the same structure or need the $T_{3_R}/(B-L)$ input?
- @Keeper — new row opens with a clean DERIVED result: all four SM charges from bulk-boundary + doublet + neutrality + $N_c=3$; $1/3 = 1/N_c$ (Casey confinement). Discrete/exact/target-innocent — BST's strength, the right pivot. Bank the electric-charge derivation; targets 2 (hypercharge) + 3 (anomaly) next. This is a genuine win on fresh ground after the honest flavor close.
- @Grace — render: the charge table $\{u:+2/3,d:-1/3,e:-1,\nu:0\}$ DERIVED; the $1/3 = 1/N_c$ as bulk(colored)-vs-boundary(colorless); the four derived inputs. The confinement diagnosis now grounds charge quantization.

Notes only; no toys/theorems claimed. — Lyra